Test on Strings

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Time: 60 mins
Full marks: 5*10 = 50
(Attempt any 5 questions out of 6)

1. String Compression Program
Write a Java program to accept a string and compress it by replacing consecutive repeated characters with the character followed by its count.
Example:
Input: aaabbccccd
Output: a3b2c4d1
Requirements:
- Handle both uppercase and lowercase separately.
- If no repetition, still display count as 1.

2. Palindrome Word Extractor
Write a program to input a sentence and display all palindrome words present in it.
Example:
Input: Madam Arora teaches malayalam and level stats
Output: Madam Arora malayalam level stats
Requirements:
- Ignore case while checking.
- Remove punctuation if present.
- Words should be printed in original format.

3. Frequency of Each Character (Without Using

Arrays)
Write a program to input a string and display the frequency of each character present in the string.
Example:
Input: programming
Output:
p - 1
r - 2
o - 1
g - 2
a - 1
m - 2
i - 1
n - 1
Constraints:
- Use only string functions and loops.
- Avoid printing duplicate entries.

4. Longest Word Finder
Write a program to input a sentence and find the longest word in it.
Example:
Input: Java programming is powerful and versatile
Output: programming
Additional Requirements:
- If multiple words have same maximum length, print the first one.
- Ignore punctuation marks.

5. String Rotation Check
Write a Java program to check whether one string is a rotation of another string.

Example:
Input:
String 1: ABCD
String 2: CDAB
Output: Rotation
Requirements:

- Logic must be implemented manually using string operations.
- Return "Not Rotation" if condition fails.

6. Custom Encryption Program
Write a program to encrypt a string based on the following rules:

- Replace each vowel with the next character.
- Replace each consonant with the previous character.
- Digits remain unchanged.

Example:
Input: Satyender123
Output: Rbuxdmcdo123
Explanation:

- S → R (previous consonant)
- a → b (next vowel)
- t → s (previous consonant)

Requirements:

- Maintain case sensitivity.
- Handle both uppercase and lowercase separately.